

CS & IT ENGINEERING



THEORY OF COMPUTATION

Regular language.

Lecture No. - 03



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Recap of Previous Lecture



Topic

Regular Expression

Topic

Construction of Regular Expression

Topic

DFA States

{{ Regular Language Detection }}

Topics to be Covered



Topic

Closure Properties of Regular Language

Topic

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Topic

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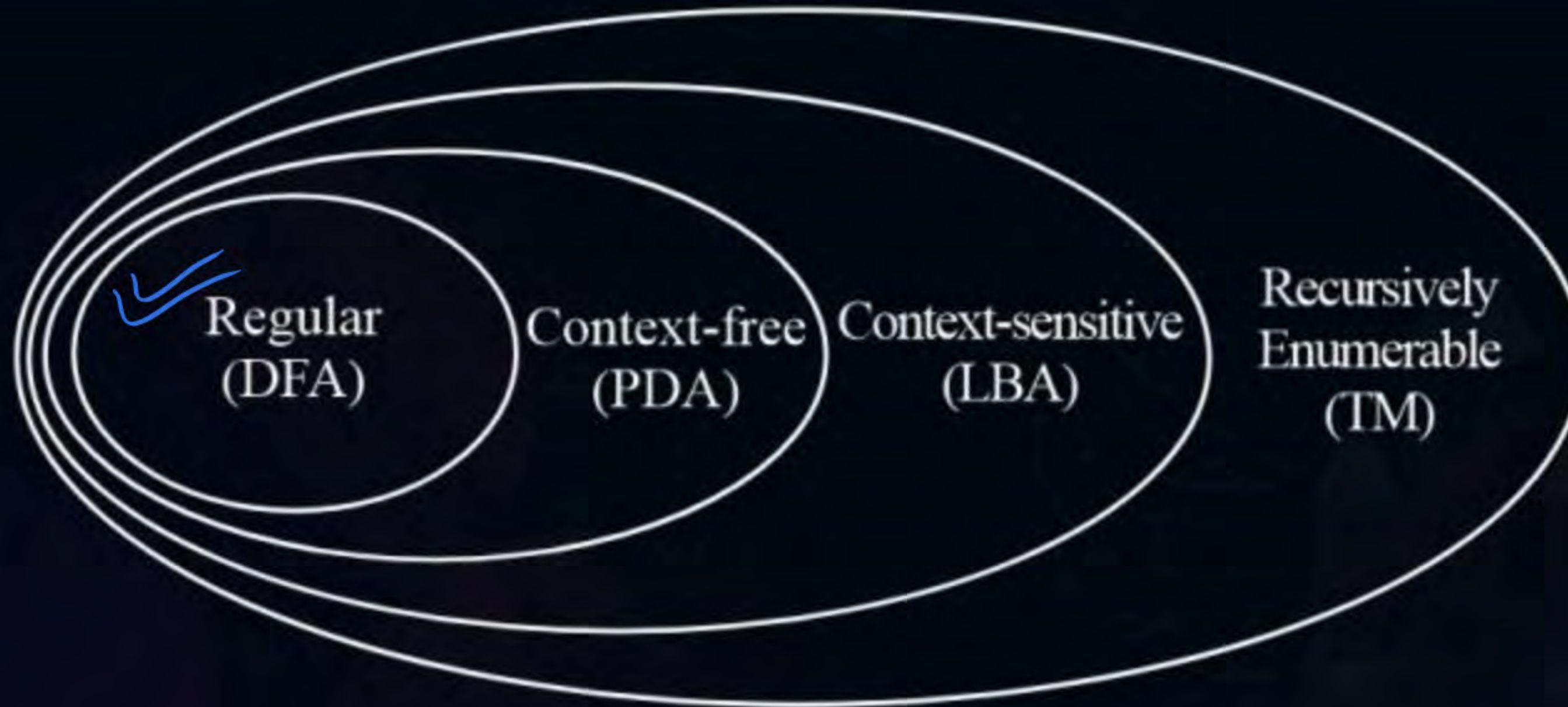
Union op

Complement





Topic : Theory of Computation



① Subset Operation :-

$$\{A \subseteq B\}$$

[not closed]

Regular

$$(a^*b^*)$$

$$\subseteq (a+b)^*$$

Subset of Regular

language is
may (or) may not be

Regular.

$$(a^n b^n)$$

$$\subseteq (a+b)^*$$

Non Regular

GATE

(Q) Which of the following is true?

(a) Subset of a regular set is always regular

(b) Subset of non regular set $\{a^n b^n\} \subseteq \{a^n b^n c^m\}$ is always regular

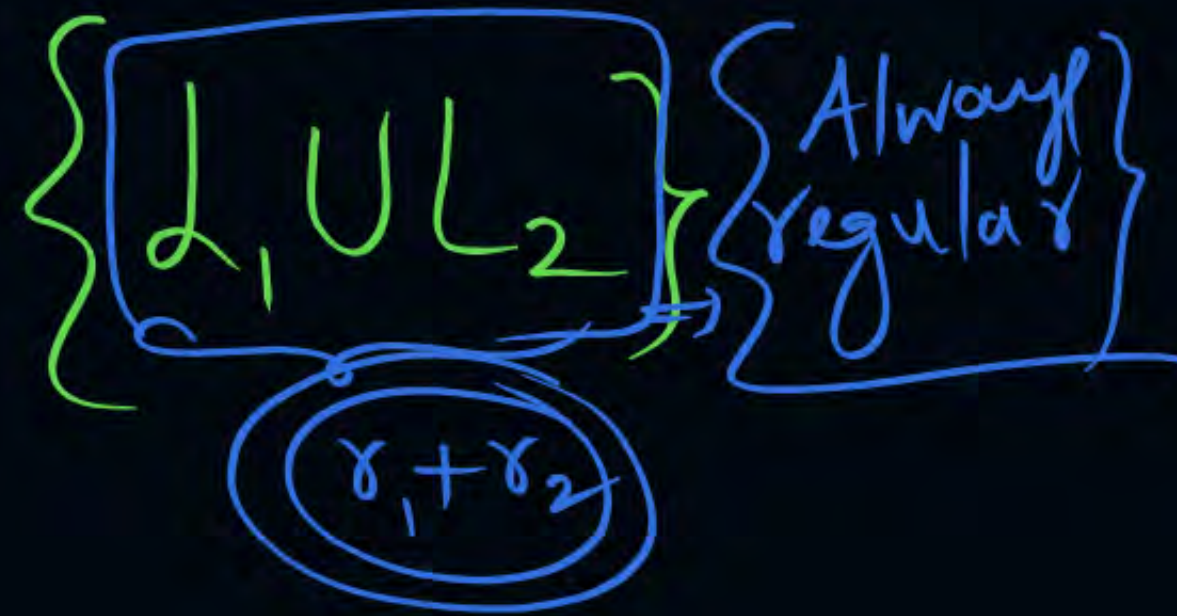
(c) Subset of infinite set is always regular

☒ (d) Subset of finite set is always regular

② Union op

closed

L_1, L_2 Regular Language



Concatenation op

reg $L_1 = \{a^n\} \quad a^*$

reg $L_2 = \{b^n\} \quad b^*$

$$\left\{ \underline{L_1 \cdot L_2} = \{a^*b^*\} \text{ Always } \underline{\text{Regular}} \right\}$$

Operation

RegLang



⑧ Difference op $[L_1 - L_2 = L_1 \setminus L_2]$ $\longrightarrow$ ✓

⑨ Reversal op $[L^R]$ $\longrightarrow$ ✓

⑩ Infinite Union op $\longrightarrow$ ✗

⑪ Infinite Intersection op $\longrightarrow$ ✗

⑫ Prefix op $\longrightarrow$ ✓

⑬ Suffix op $\longrightarrow$ ✓

Op

Reg. Lang

⑭ Substitution op



⑮ Homomorphism op



⑯ Inverse Homomorphism op



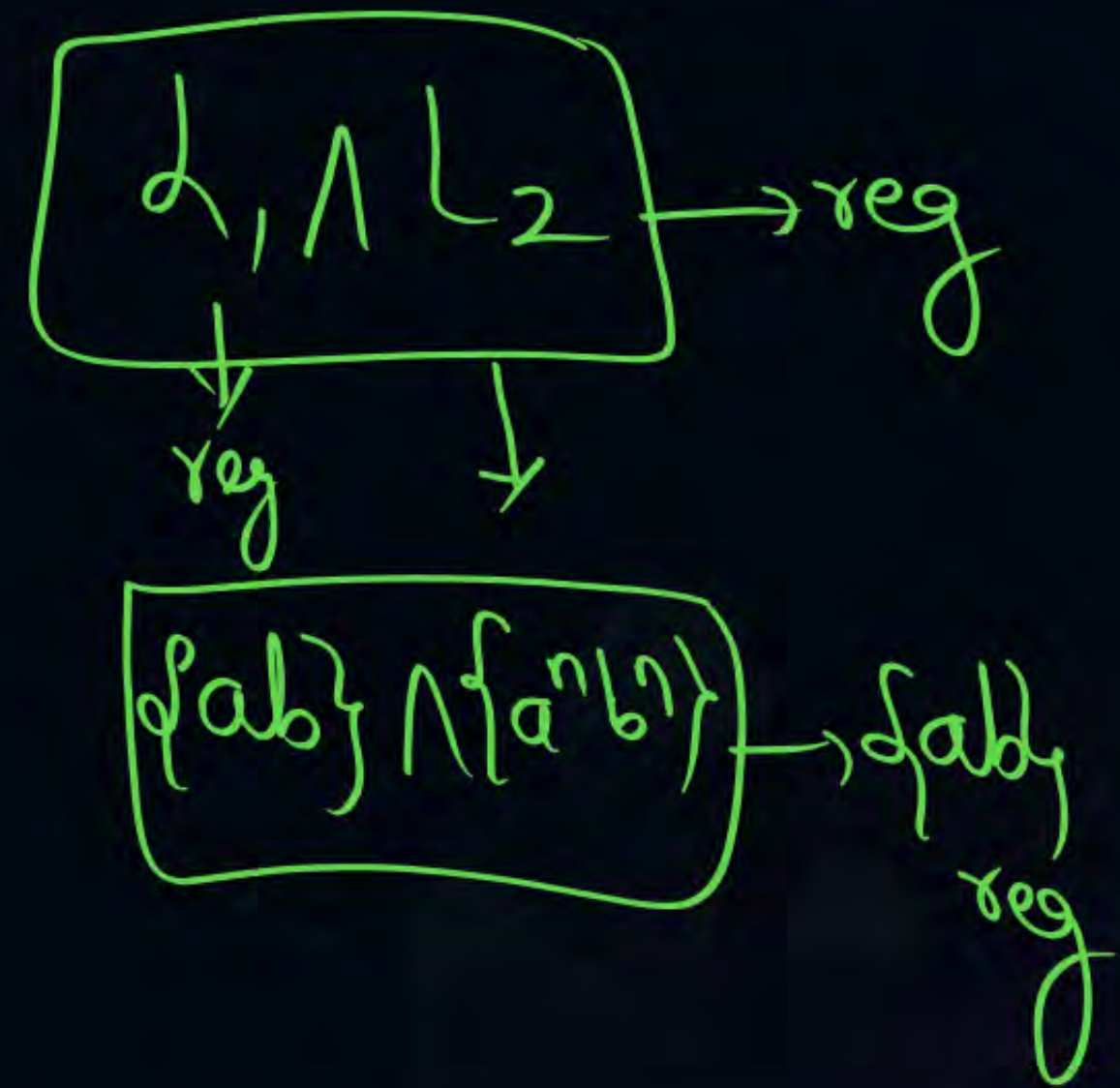
(Q) If $L_1 \cap L_2$ is always regular and L_1 is regular then L_2 is —

(a) always regular

(b) need not be regular

(c) always non regular

(d) none



Substitution op

$$\Sigma = \{0, 1\}$$

$$\Delta = \{a, b\}$$

$$S(0) = \underline{(a+b)(a+b)}$$

$$S(1) = [(a+b)^2]^*$$

$$L = 0^* = \text{Diagram 1} \Rightarrow \text{Diagram 2} \quad (a+b)(a+b)$$

$$S(L) = S(0^*) = [(a+b)(a+b)]^* = \text{Diagram 3}$$

$$[S(0)]^*$$

$$S(L) = [(a+b)(a+b)]^*$$

Homomorphism

$$\Sigma = \{0, 1\}$$

$$\Delta = \{a, b\}$$

$$h(0) = aa$$

$$h(1) = \underline{\underline{bb}}$$

$$L = (0)^*$$

$$h(L) = \underline{\underline{(aa)^*}}$$

Inverse H

$$L = (aa)^*$$

$$h^{-1}(L) = \downarrow (0)^*$$

Prefix op:-
TOC

$\left\{ \begin{array}{l} \epsilon \\ T \\ \hline TO \\ \hline TOC \end{array} \right\}$

Sequence of leading symbols over the given string.

gate

$\left\{ \begin{array}{l} \epsilon \\ g \\ ga \\ gat \\ gate \end{array} \right\}$

delhi

$\left\{ \begin{array}{l} \epsilon \\ d \\ de \\ del \\ delh \\ delhi \end{array} \right\}$

abc --- n



n+1

total prefixes

Suffix of string :-

~~→~~ ~~→~~ ~~→~~
T O C

{
ε
c
oc
TOC
}

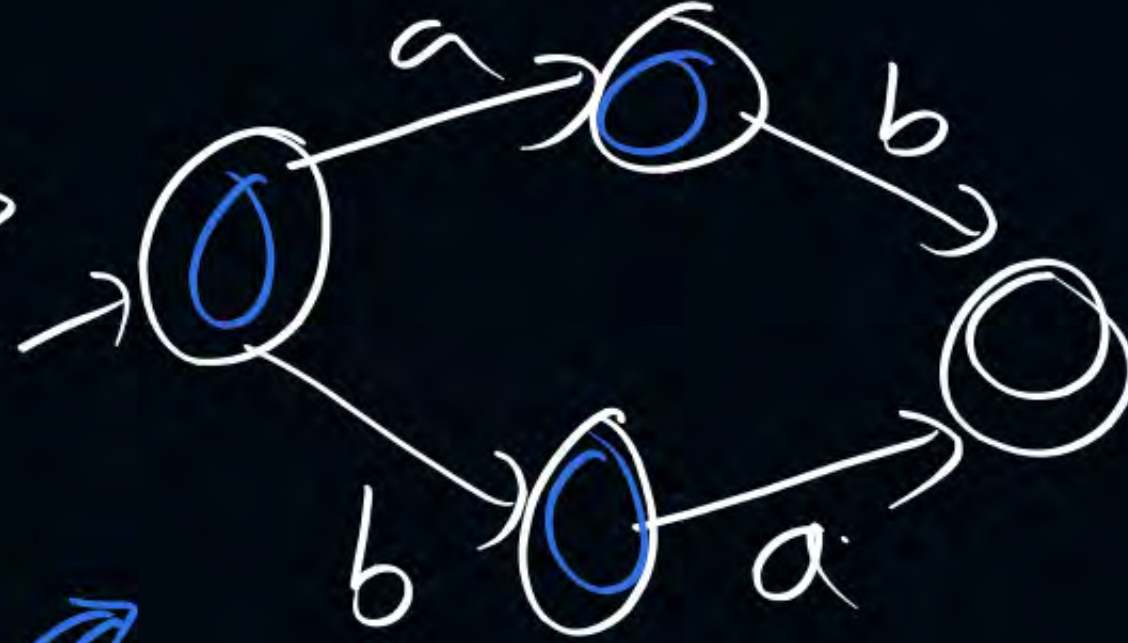
gate

{
ε
e
te
ate
gate
}

abc...n

(n+1) {Suffixes}

$$L = \{ab, ba\} \Rightarrow$$



$$\text{prefix}(L) = \{ \epsilon, a, b, ab, ba \}$$

$$= \{ \epsilon, a, b, ab, ba \}$$

Infinite Union
not closed

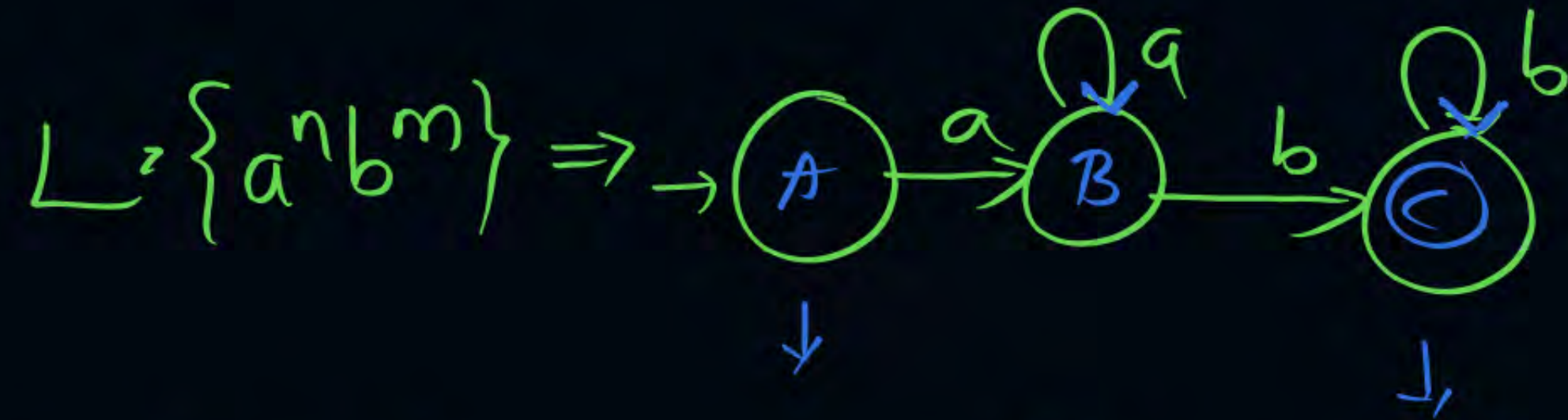
$L_1, L_2, L_3, \dots$ regular

$$L_1 \cup L_2 \cup L_3 \cup L_4 \cup \dots$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$\{ab\} \cup \{a^2b^2\} \cup \{a^3b^3\} \cup \{a^4b^4\} \cup \dots = \boxed{\{a^n b^n \mid n \geq 1\}}$$

Reg language not closed for Infinite Union
But closed for finite Union



$R = \{b^m a^n\}$



Consider the languages $L_1 = \phi$ and $L_2 = \{a\}$. Which one of the following represents $L_1 L_2^* \cup L_1^*$?

[2013: 1 Mark]

$$\phi \cdot a^* + \epsilon$$

$$\phi + \epsilon = \epsilon$$



- ☒ A $\{\epsilon\}$
- ☐ B ϕ
- ☐ C a^*
- ☐ D $\{\epsilon, a\}$

THANK - YOU